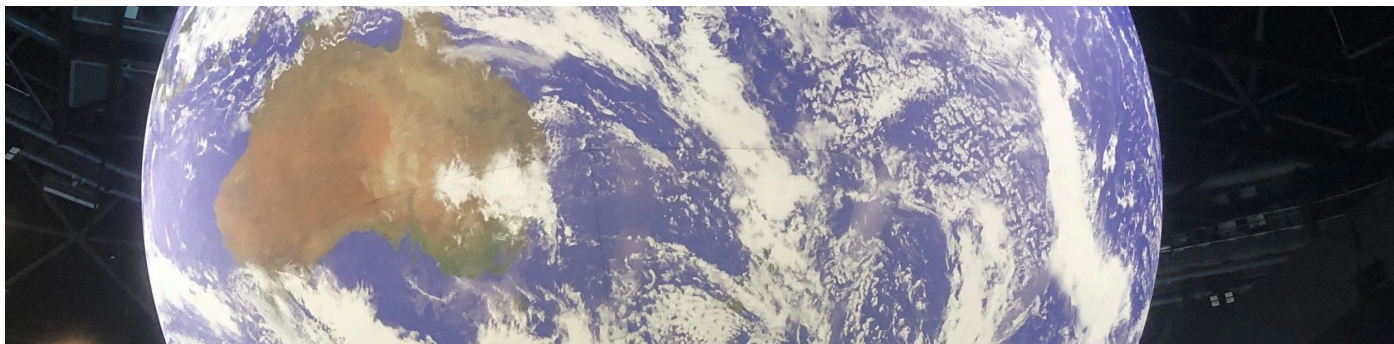


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Navigating Global Climate Change: Science, Emissions, and International Policy

Abstract

We are all passengers on spaceship Earth, but there is no captain. After the COP26 climate summit in 2021, climate scientist Kevin Trenberth reflected on what the latest science says, why international cooperation has fallen short, and what role a small country like New Zealand can play in slowing climate change.

Introduction

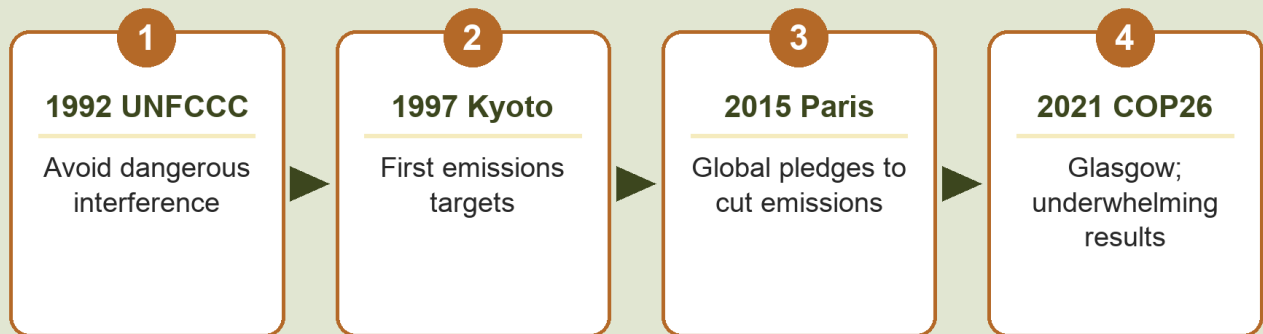
The Intergovernmental Panel on Climate Change reported in 2021 that recent changes in the climate are widespread, rapid, intensifying, and unprecedented in thousands of years, and that it is indisputable that human activities are the cause. But why is climate change so harmful when people live comfortably in climates from the tropics to the Arctic? Trenberth explains that it is the change itself that is disruptive: it alters what we are adapted to, especially farming and food, and it makes extreme weather worse. As a scientist who has played a major role in past IPCC assessments, he asks how the world, and New Zealand, can respond.



How Do Scientists Study This?

Trenberth combined three kinds of analysis. First, he summarized the physical science: how greenhouse gases heat the planet, and how extra heat increases drying where it is dry and heavier rain where it rains. Second, he reviewed the history of international climate cooperation, from the 1992 UN Framework Convention on Climate Change through the Kyoto Protocol, the 2015 Paris Agreement, and COP26 in Glasgow. Third, he examined New Zealand's unusual emissions profile, with as much from methane as from carbon dioxide, and compared its approach to that of other countries, especially Australia's fugitive methane emissions.

The road of international climate policy



Review the science

Summarize how greenhouse gases heat the planet and intensify extremes.

Trace the treaties

Follow international climate agreements from 1992 to COP26.

Assess one country

Examine New Zealand's emissions, commitments, and options for action.

What Did They Discover?

Global temperature has risen 1.1°C since pre-industrial times. Extra heat dries land and fuels storms that can produce 30% more rain. By about 2°C, expected in the 2050s, extreme weather will be common, crops may fail, water shortages will spread, and environmental refugees could fuel conflicts. Yet international governance is weak: the UN needs unanimous consensus, and COP26's results were underwhelming. New Zealand's methane, mostly from farming, lasts about nine years, so cuts pay off quickly; but other countries focus on fossil fuel leaks, and Australia's fugitive methane exceeds all of New Zealand's methane emissions. Trenberth argued New Zealand relies too heavily on offsets and should cut coal, expand wind, solar, and hydro, and decarbonize.

Climate change by the numbers

1.1°C global warming since pre-industrial times

2050s when 2°C of warming is expected

~9 yrs average lifetime of methane in the atmosphere

NASA launched the DART mission to test deflecting an incoming asteroid. Trenberth asks: where is the spaceship to bump us back on course with climate change?

Takeaways

Change is the danger

Climate change disrupts what farms, cities, and ecosystems are adapted to.

No captain

International agreements depend on consensus and have not done enough.

Lead by example

Small countries can cut emissions and support vulnerable neighbours.

Key Terms

UNFCCC

The United Nations Framework Convention on Climate Change, a 1992 treaty to prevent dangerous human interference with the climate.

COP

The Conference of the Parties, the annual UN climate meeting where countries negotiate climate action.

Kyoto Protocol

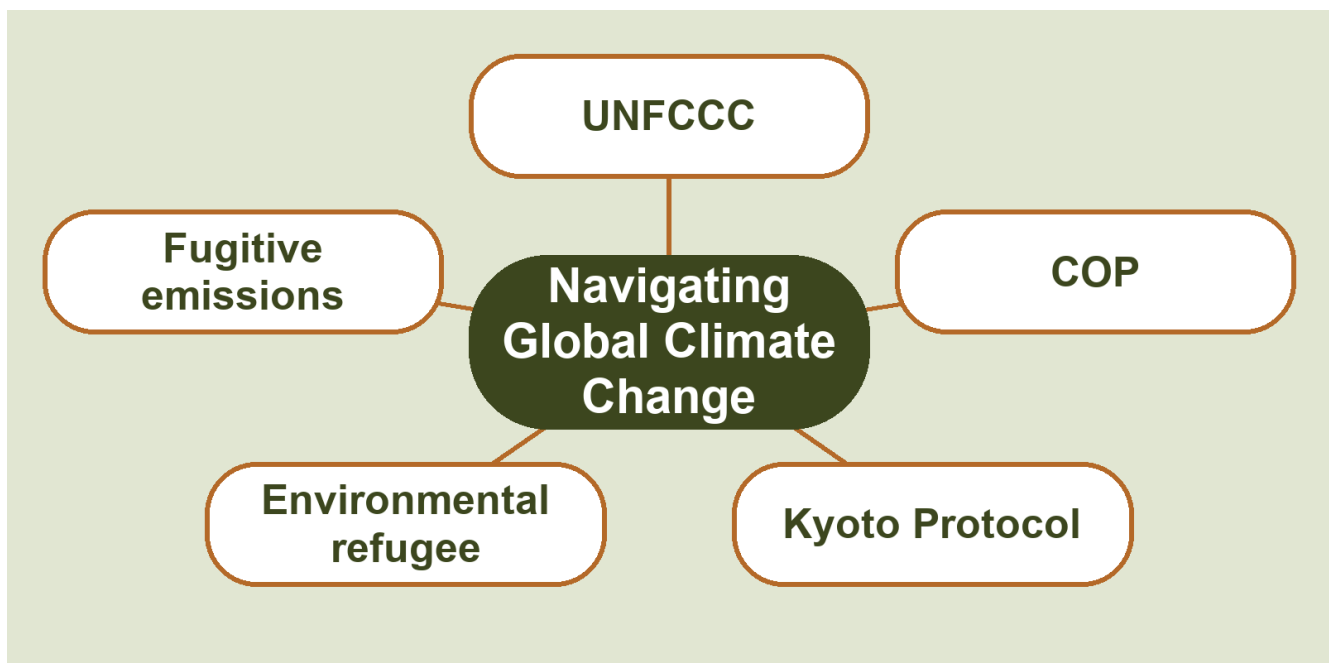
A 1997 international agreement that set emissions targets for developed countries.

Environmental refugee

A person forced to leave home because of environmental changes such as drought, floods, or sea-level rise.

Fugitive emissions

Gases, such as methane, that leak unintentionally from fossil fuel operations like pipelines and wells.



Activity

Spaceship Earth Council

Students simulate a climate negotiation to experience the challenge of reaching global agreement.

Materials: Country role cards (large emitter, small island nation, farming nation, developing economy); Agreement template; Timer

1. Each group represents a country and reads its role card, including its emissions and vulnerabilities.
2. Propose one commitment your country is willing to make and one it wants from others.
3. Negotiate a shared agreement. Rule: every country must agree, just like the UN.
4. Reflect: what made consensus hard? What compromises were made?
5. Write a short speech for New Zealand's delegate explaining how a small country can lead by example.

References

Trenberth, K. Climate: NZ's role on spaceship Earth. Newsroom.
<https://www.newsroom.co.nz/whats-nzs-role-on-spaceship-earth>